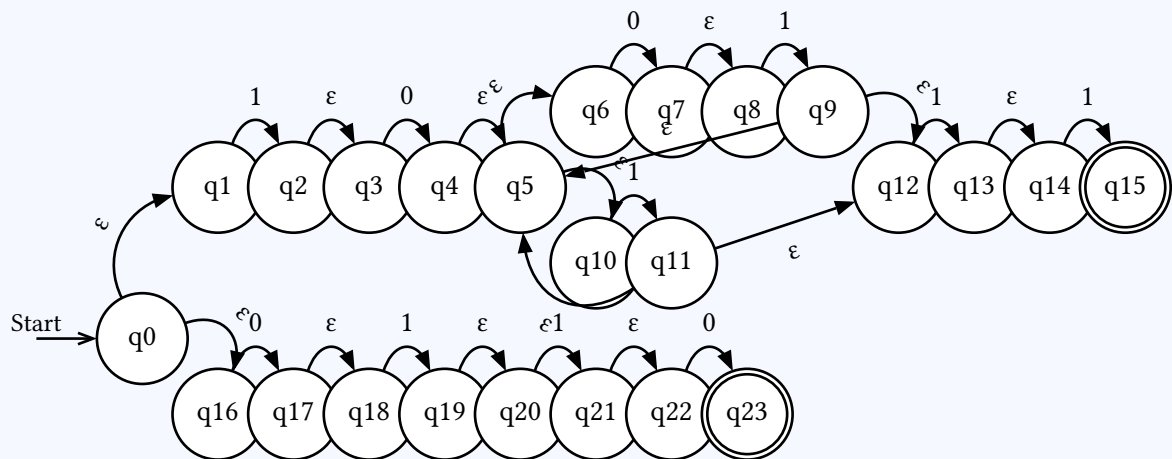


Contents

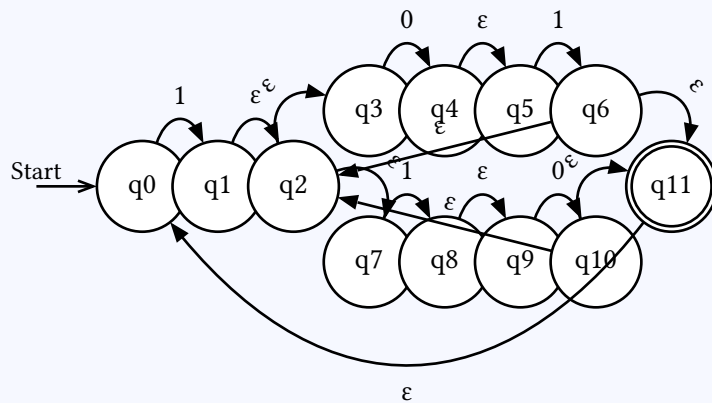
1. The Great Expression 1
2. Puedo ir(regular) al bano 2
3. Count Yourself Blist! 4

Question 1. The Great Expression

(a)



(b)



Question 2. Puedo ir(regular) al bano

(a)

Suppose, for the sake of contradiction, that S is regular. Then there is a DFA M such that M accepts exactly S .

Let $R = \{(^n : n \geq 0\}$. Because the DFA is finite and R is infinite, there are two different strings x, y in R such that x and y go to the same state when read by M . Since both are in R , $x = (a$ for some integer $a \geq 0$, and $y = (b$ for some integer $b \geq 0$, with $a \neq b$.

Consider the string $z =)^a$.

Since x, y led to the same state and M is deterministic, xz and yz will also lead to the same state q in M . Observe that $xz = (a)^a$, so $xz \in S$, but $yz = (a)^b$, so $yz \notin S$. Since q can only be either an accept or a reject state, but not both, M does not actually recognise S . That's a contradiction!

Therefore, S is an irregular language.

(b)

Suppose, for the sake of contradiction, that $L = \{0^m 1^n : m, n \in \mathbb{N} \text{ and } m \mid n\}$ is regular. Then there is a DFA M such that M accepts exactly L .

Let $S = \{0^n : n \in \mathbb{N} \text{ and } n \text{ is prime}\}$. Since there are infinitely many prime numbers, and because the DFA is finite and S is infinite, there are two different strings x, y in S such that x and y go to the same state when read by M . Since both are in S , $x = 0^a$ for some prime integer a , and $y = 0^b$ for some prime integer b .

Consider the string $z = 1^a$.

Since x, y led to the same state and M is deterministic, xz and yz will also lead to the same state q in M . Observe that $xz = 0^a 1^a$, and since a divides a , $xz \in L$. However, $yz = 0^b 1^a$, and since a is prime and can only be divided by itself and 1, and $b \neq 1$ since b is prime, $yz \notin L$. Since q can only be either an accept or a reject state, but not both, M does not actually recognise L . That's a contradiction!

Therefore, L is an irregular language.

Continued on next page $\rightarrow$

(c)

Suppose, for the sake of contradiction, that $L = \{0^m 1^n : m, n \in \mathbb{N} \text{ and } m \leq n^2\}$ is regular. Then there is a DFA M such that M accepts exactly L .

Let $S = \{0^n : n \in \mathbb{N} \text{ and } n > 0 \text{ is a square number}\}$. Since there are infinitely many square numbers, and because the DFA is finite and S is infinite, there are two different strings x, y in S such that x and y go to the same state when read by M . Since both are in S , $x = 0^a$ for some square number $a > 0$, and $y = 0^b$ for some square number $b > 0$.

Note that since $a \neq b$, $a > b$ or $b > a$. We proceed by cases...

Suppose that $a > b$. Consider the string $z = 1^{\sqrt{b}}$. Note that this is valid, since b is a square number.

Since x, y led to the same state and M is deterministic, xz and yz will also lead to the same state q in M . Observe that $yz = 0^b 1^{\sqrt{b}}$, and since $b = \sqrt{b}^2$, $yz \in L$. However, $xz = 0^a 1^{\sqrt{b}}$, and since $a > \sqrt{b}^2$, $xz \notin L$. Since q can only be either an accept or a reject state, but not both, M does not actually recognise L . That's a contradiction! Therefore, if $a > b$, then L is an irregular language.

Now suppose that $b > a$. Consider the string $z = 1^{\sqrt{a}}$. Note that this is valid, since a is a square number.

Since x, y led to the same state and M is deterministic, xz and yz will also lead to the same state q in M . Observe that $xz = 0^a 1^{\sqrt{a}}$, and since $a = \sqrt{a}^2$, $xz \in L$. However, $yz = 0^b 1^{\sqrt{a}}$, and since $b > \sqrt{a}^2$, $yz \notin L$. Since q can only be either an accept or a reject state, but not both, M does not actually recognise L . That's a contradiction! Therefore, if $b > a$, then L is an irregular language.

Since our cases were exhaustive, we have shown that L is an irregular language.

Question 3. Count Yourself Blist!

(a)

We will use the dovetailing method to show that A is countable.

- i. Each element $L \in A$ can be assigned to group N , where $N = \text{len}(L)$.
- ii. Since $N \in \mathbb{N}$, we can use the natural ordering of numbers to order these groups.
- iii. Within each group N , for each element L , let B be $\sum_{i=0}^N L[i] \cdot 2^i$, essentially converting each list to a binary encoding. Since B is a unique finite number, we can order the elements of N by the natural ordering of numbers.

Therefore, we have shown, by dovetailing, that A is countable.

(b)

Consider an arbitrary element $L \in B$. L is also an element of A (from the previous solution), since A contains **all** lists whose elements are in $\{0, 1\}$.

Since L was arbitrary, by definition of subset, B is a subset of the countable set A .

Thus, B must be countable, because it is a subset of a countable set.

(c)

Suppose, for the sake of contradiction, that C is countable.

Then, there exists a surjection (onto function) $f : \mathbb{N} \rightarrow C$. So for every natural number i , we have some element C_i that i maps to.

We now construct an element x of S such that $x[i] = \sim C_i[i]$. That is, $x[i]$ is 1 if $C_i[i]$ is 0, and $x[i]$ is 0 if $C_i[i]$ is 1.

So, no $f(i)$ maps to the element x of C . But then f is not a surjection and there exists no way for us to enumerate all elements of C . That's a contradiction!

Therefore, C is uncountable.